CHIRUNHO

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS 9188/1

PAPER 1 Multiple Choice

Tuesday **4 NOVEMBER 2003** Afternoon 1 hour

Additional materials:
- Electronic calculator and/or Mathematical tables
- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (Type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

10mm = 1cm
100cm = 1000m
1m = 1000mm
0/1

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

[Turn over

Data

speed of light in free space,	$c = 3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$
permeability of free space,	$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$
permittivity of free space,	$\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F\,m^{-1}}$
elementary charge,	$e = 1.60 \times 10^{-19}\ \mathrm{C}$
the Planck constant,	$h = 6.63 \times 10^{-34}\ \mathrm{J\,s}$
unified atomic mass constant,	$u = 1.66 \times 10^{-27}\ \mathrm{kg}$
rest mass of electron,	$m_e = 9.11 \times 10^{-31}\ \mathrm{kg}$
rest mass of proton,	$m_p = 1.67 \times 10^{-27}\ \mathrm{kg}$
molar gas constant,	$R = 8.31\ \mathrm{J\,K^{-1}\,mol^{-1}}$
the Avogadro constant,	$N_A = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$
the Boltzmann constant,	$k = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
gravitational constant,	$G = 6.67 \times 10^{-11}\ \mathrm{N\,m^{2}\,kg^{-2}}$
acceleration of free fall,	$g = 9.81\ \mathrm{m\,s^{-2}}$

Formulae

uniformly accelerated motion,	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas,	$W = p\Delta V$
gravitational potential,	$\phi = -\dfrac{Gm}{r}$
refractive index,	$n = \dfrac{1}{\sin C}$
resistors in series,	$R = R_1 + R_2 + \ldots$
resistors in parallel,	$1/R = 1/R_1 + 1/R_2 + \ldots$
electric potential,	$V = \dfrac{Q}{4\pi\epsilon_0 r}$
capacitors in series,	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel,	$C = C_1 + C_2 + \ldots$
energy of charged capacitor,	$W = \frac{1}{2}QV$
alternating current/voltage,	$x = x_0 \sin \omega t$
hydrostatic pressure,	$p = \rho g h$
pressure of an ideal gas,	$p = \frac{1}{3}\dfrac{Nm}{V}\langle c^2 \rangle$
radioactive decay,	$x = x_0 \exp(-\lambda t)$
decay constant,	$\lambda = \dfrac{0.693}{t_{\frac{1}{2}}}$
critical density of matter in the Universe,	$\rho_0 = \dfrac{3H_0^2}{8\pi G}$
equation of continuity,	$Av = \text{constant}$
Bernoulli equation (simplified),	$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$
Stokes' law,	$F = Ar\eta v$
Reynolds' number,	$R_e = \dfrac{\rho v r}{\eta}$
drag force in turbulent flow,	$F = Br^2\rho v^2$

[Turn over

1 The relationship between four physical quantities x, p, q and t is given by $x = p + qt$, where t is the time in seconds. If q has units of ms^{-1}, what are the units of p?

A ms^{-1}
B ms^{-2}
C ms
D m

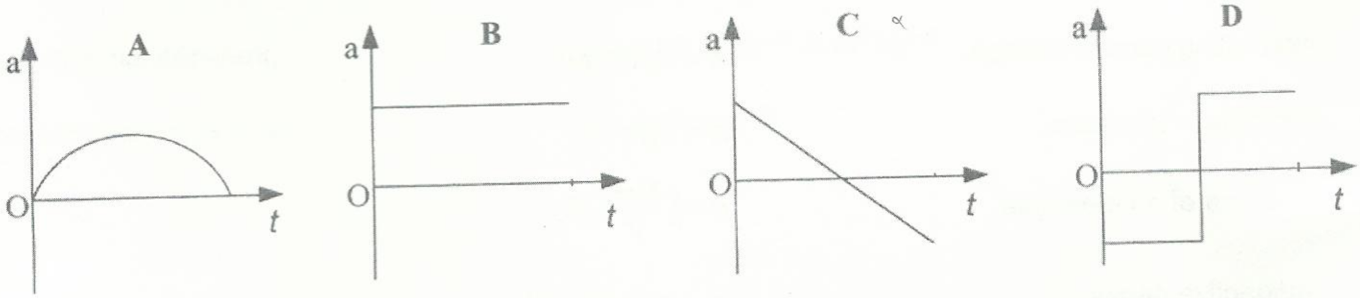
$ms^{-1}s^{1}$

2 In an experiment, the external diameter d_1 and the internal diameter d_2 of a metal tube are found to be (64 ± 2)mm and (47 ± 2)mm respectively. What is the expected percentage error in $(d_1 - d_2)$ from these readings?

A 1%
B 6%
C 12%
D 24%

3,125

3 A ball is thrown vertically upwards. Which of the following graphs shows how the acceleration '*a*' varies with time '*t*'?



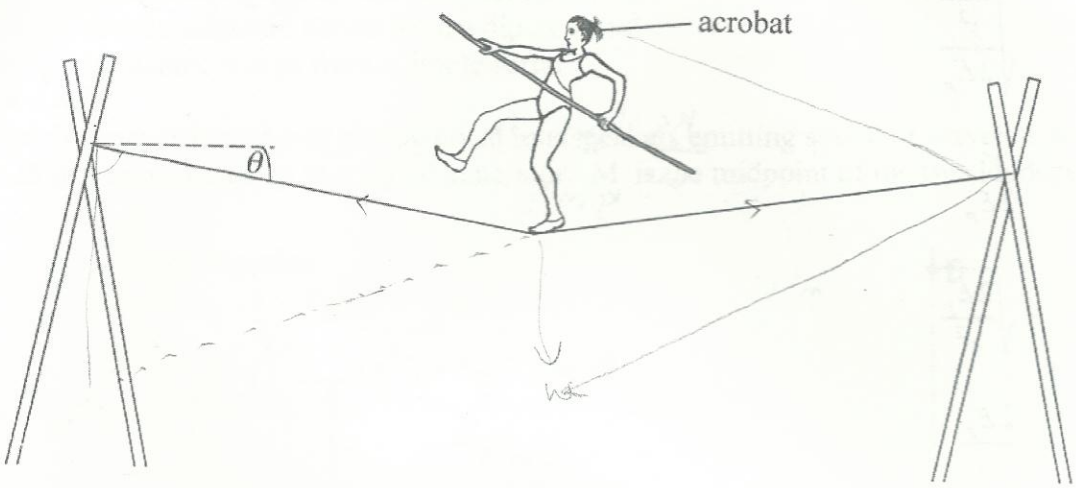
4 Which of the following has the same units as momentum?

A force
B pressure
C the product of force and time
D product of force and velocity

mv

$kgms^{-1}$

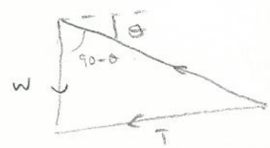
5 An acrobat is stationary at the centre of a tight rope. The angle θ between the tight rope and the horizontal is as shown below.



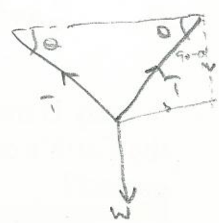
If the acrobat has a weight W, what is the tension in the rope?

A $\frac{W}{2}$

B $\frac{W}{2\sin\theta}$



C $\frac{W}{2\cos\theta}$ $\qquad \frac{T}{\sin(90-\theta)} = W$



D $\frac{W}{\sin\theta}$

6 A truck with only the driver in it, moving with a velocity u, can be stopped by applying brakes in a distance s. When the truck was loaded its total mass increased by 30%. The driver applied the same braking force to stop the truck while it was moving with the velocity u. What was the new braking distance? $180-2\theta$

A $0.3s$
B $0.7s$
C $1.0s$
D $1.3s$

$S = ut \quad 0 = u^2 + 2as.$

$F = ma. \quad F_1 = ma = 1.3ma,$

$F_1 =$

$v^2 = u^2 - 2as \qquad a_r = \frac{a}{1.3}$

$S = ut - \frac{1}{2}at^2.$

$\frac{u^2}{2a} = 2a(s).$

[Turn over

7 A body experiences free fall from rest. It has a velocity v when it has lost the amount of gravitational potential energy E_p. What is the mass of the body?

kg m² s⁻²

Nm² kg⁻².

A $\sqrt{\dfrac{v^2}{2E_p}}$

kg kgms⁻² m² kg⁻² kg. −GM/r.

kg m² s⁻² / v²

B $\dfrac{v^2}{2E_p}$

m² s⁻²

N m² kg⁻² m² s⁻²

kg m s⁻² m² kg⁻² kg

m² s⁻²

m² s⁻² / N m² kg⁻²

kg m s⁻² m² kg⁻²

C $\sqrt{\dfrac{2E_p}{v^2}}$

D $\dfrac{2E_p}{v^2}$

8 A body of mass m moves at constant angular speed ω in a horizontal circle of radius r. What is the work done on the body in one revolution?

F = mω²r = mv²/r

A zero

B $2\pi m r^2 \omega^2$

C $m r^2 \omega^2$

D $\dfrac{1}{2} m r^2 \omega^2$

9 A body is moved from point **P** on the Earth's surface to another point **Q** further away from the Earth's centre. Which statement about the gravitational potential at the two points is correct?

A It is positive at both points and greater at **Q** than at **P**.

B It is positive at both points and less at **Q** than at **P**.

C It is negative at both points and greater at **Q** than at **P**.

D It is negative at both points and less at **Q** than at **P**.

P Q

10 A body in simple harmonic motion makes n complete oscillations per second. What is the angular frequency of this motion?

f = n s⁻¹ T = 1/n.

ω = 2π/T = 2π/(1/n)

A n rad s^{-1}

B $\dfrac{1}{n}$ rad s^{-1}

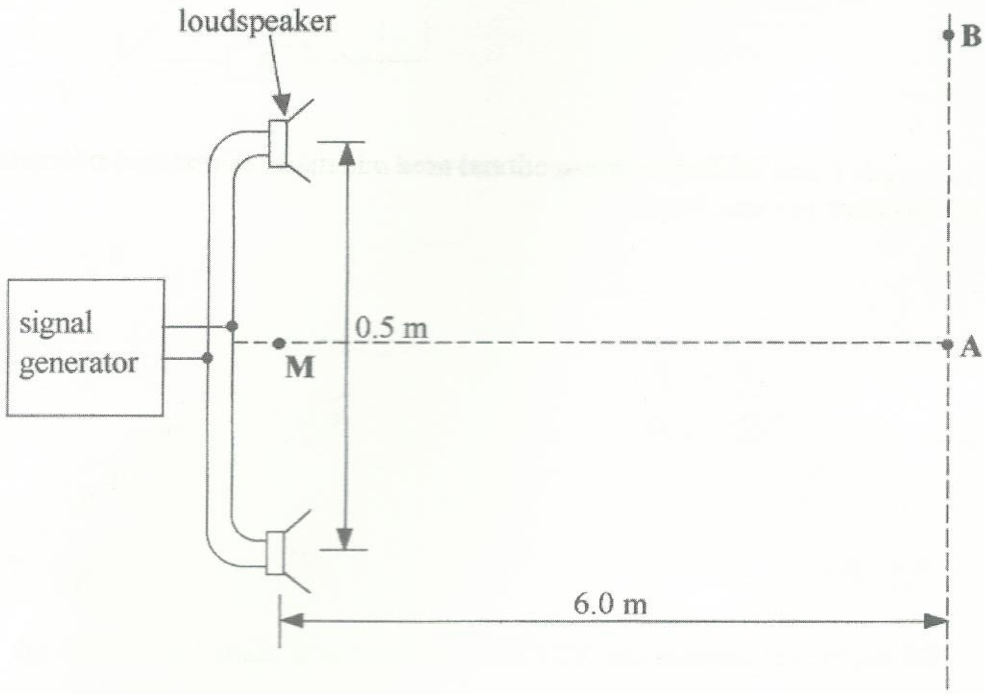
C $\dfrac{2\pi}{n}$ rad s^{-1}

D $2\pi n$ rad s^{-1}

11 In which instance are the waves plane polarised?

A infrared radiation from a hot electric iron
B compression waves caused by an earthquake
C electromagnetic waves from a dipole aerial
D ultrasonic waves from a dipole aerial

12 The diagram below shows two identical loudspeakers emitting sound of wavelength 0.28 m when connected to a signal generator. M is the midpoint of the two loudspeakers.

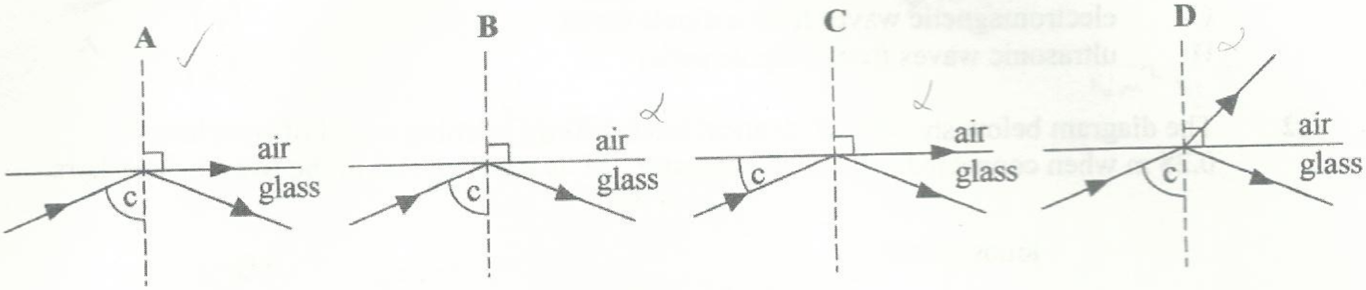


A student moving along line **AB** hears a loud sound at **A** which gradually reduces until virtually nothing is heard at **B**. What is the distance between **A** and **B**?

A 0.25m
B 0.50m
C 1.68m
D 3.36m

$\lambda = \frac{a\Delta x}{D}$ $\Delta x = \frac{(0{,}28)(6)}{0{,}5}$

13 A ray of light is incident on a glass–air boundary for which the critical angle is c. Which diagram is correct?



14 A conductor of length ℓ and uniform cross-sectional area a is made of material of resistivity ρ. What is its resistance per unit length?

A $\dfrac{\rho}{a}$

B $\dfrac{a}{\rho}$

C $\rho\dfrac{\ell}{a}$

D $\dfrac{a}{\rho\ell}$

$\frac{R}{m} = \frac{\rho l}{a}$

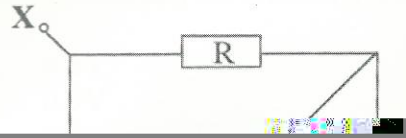
15 A generator produces 100kW of power at a potential difference of 10kV. The power is transmitted through cables of total resistance 5Ω. What is the power loss in the cables?

A 0.05W
B 50W
C 500W
D 2×10^7W

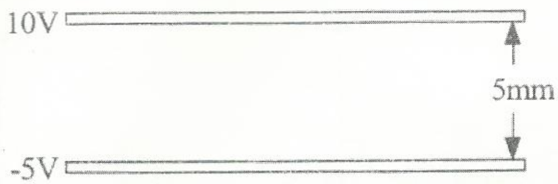
I^2R

$P = VI$

16 The diagram shows a network of resistors each of resistance R.



18 The top plate of a parallel plate capacitor is at 10V and the bottom one is at – 5V as shown in the diagram.

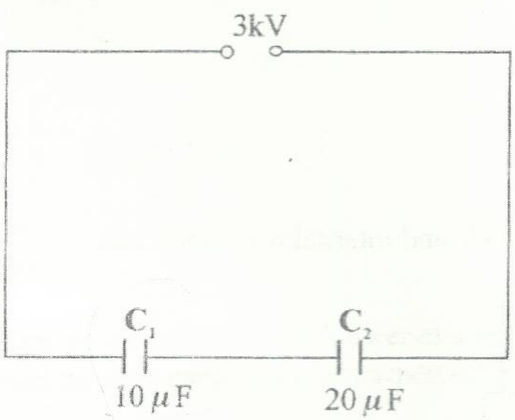


If the distance between the plates is 5mm, what is the magnitude of the electric field strength between the plates?

$E = \frac{V}{d}$

- **A** 7.5×10^{-2} Vm^{-1}
- **B** 3.0×10^{2} Vm^{-1}
- **C** 1.0×10^{3} Vm^{-1}
- **D** 3.0×10^{3} Vm^{-1}

19 Two capacitors C_1 and C_2 of magnitudes 10 μF and 20 μF respectively are connected in series across a 3kV dc supply as shown.



$C =$

$Q = CV$

What are the charges on C_1 and C_2?

	C_1	C_2
A	10mC	40mC
B	20mC	20mC
C	30mC	60mC
D	90mC	90mC

20 A vertical straight conductor **X** of length 0.5m is situated in a uniform horizontal magnetic field of strength 0.1T and is carrying a current of 4.0A. Through what angle must **X** be turned in a vertical plane so that the force on **X** is halved?

$F = BIL$. 0,2.

- **A** 30°
- **B** 45°
- **C** 60°
- **D** 90°

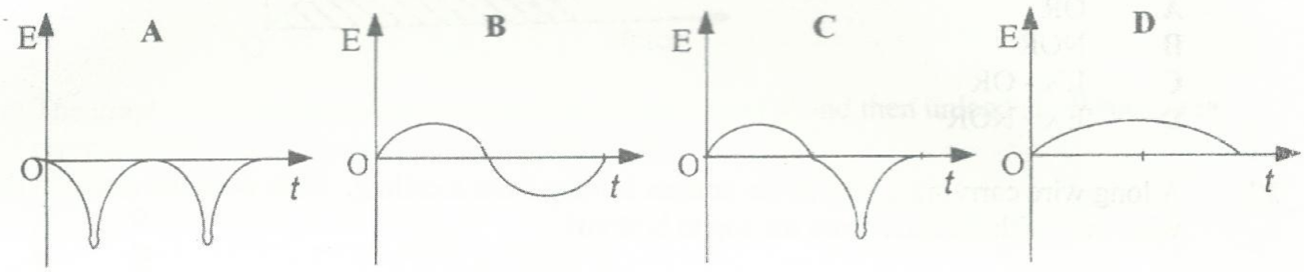
$BIL \sin\theta = 0,1$

21 An aircraft of wingspan 40m flies horizontally at a speed of $300ms^{-1}$ in an area where the vertical component of the Earth's magnetic field is 5.0×10^{-5} T. What is the e.m.f generated between the aircraft's wingtips?

E = BLV

A 0.6V
B 0.3V
C 2.4×10^{-22} V
D 0

22 A bar magnet is dropped vertically downwards through a long solenoid which is connected to an oscilloscope. Which oscilloscope trace shows how **E**, the e.m.f induced on the coil varies with the time *t* as the magnet moves downwards?



23 An alternating current of r.m.s value 2A flowing through a given resistor **R** dissipates heat at the same rate as a steady direct current **I** flowing through an identical resistor. What is the current **I**?

$I_{rms} = \frac{I_0}{\sqrt{2}}$ $(2\sqrt{2})^2 R \cdot 4R$. 8R

A 1A
B $\sqrt{2}$A
C 2A
D $2\sqrt{2}$A

24 In an ideal transformer, what is the most important function of the soft iron core?

A to increase the flux linkage between the primary and the secondary coils
B to dissipate the heat generated by the two coils
C to eliminate the back e.m.f. produced in the secondary coil
D to produce a uniform radial field in the two coils

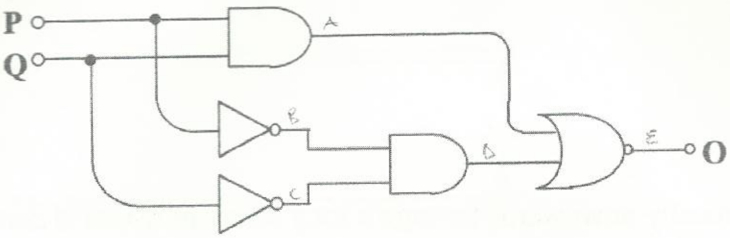
25 Which changes take place when the level of negative feedback in an amplifier in closed-loop mode is increased?

A gain and bandwidth are increased
B gain and bandwidth are reduced
C gain is increased but bandwidth is reduced
D gain is reduced but bandwidth is increased

nfb increases

P	Q	A	B	C	D	E
0	0	0	1	1	1	0
0	1	0	1	0	0	1
1	0	0	0	1	0	1
1	1	1	0	0	0	0

26 The diagram below shows a logic circuit.



Ex - OR

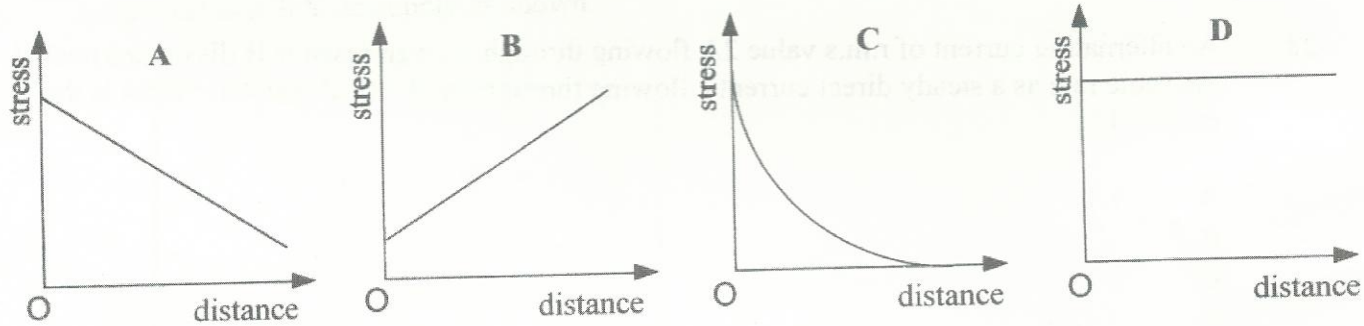
OR

NOR

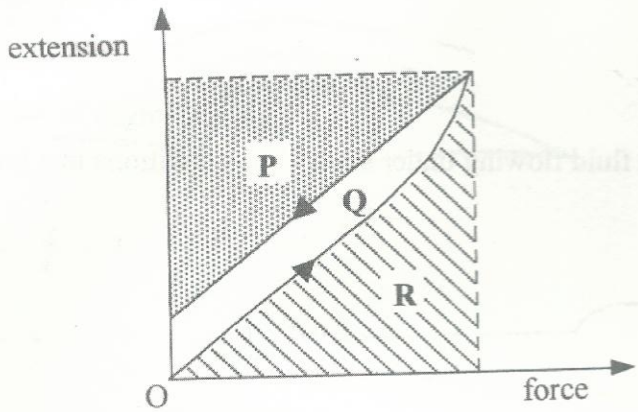
Which of the following gates has the same logic function as this circuit?

A OR
B NOR
C EX - OR
D EX - NOR

27 A long wire carrying a load at the bottom is hung from a ceiling. How does the stress in the wire vary with distance from the top to bottom?



28 The following is a graph of extension versus force for a sample of arrester wire used for halting fighter planes landing on a ship.



The graph was obtained when the wire was steadily loaded and then unloaded. Which area represents the energy recovered during unloading?

A P
B Q
C R
D Q + R

29 The temperature of a hot gas was increased by 50°C from an initial temperature of 100°C. What is the temperature change in kelvin?

A 50
B 323
C 373
D 423

~~$900 \to 150$~~

$100 \to 150$

30 The density of argon at a pressure of 1.00×10^5 Pa and at a temperature of 300 K is $1.60\ \text{kgm}^{-3}$. What is the root mean square speed of argon molecules at this temperature?

A $216\ \text{ms}^{-1}$
B $250\ \text{ms}^{-1}$
C $306\ \text{ms}^{-1}$
D $433\ \text{ms}^{-1}$

$P = \frac{1}{3}\rho c^2$

$c^2 = \frac{3P}{\rho}$

31 Five gas molecules have the speeds shown below.

speed / $\times 10^3\text{ms}^{-1}$	1.0	3.0	5.0	7.0	9.0

$1 + 9 + 25 + 49 + 81$

$165 \times (10^3)^2$

What is their root mean square speed?

A $2.2 \times 10^3\ \text{ms}^{-1}$
B $2.5 \times 10^3\ \text{ms}^{-1}$
C $5.0 \times 10^3\ \text{ms}^{-1}$
D $5.7 \times 10^3\ \text{ms}^{-1}$

32 The average kinetic energy of the molecules of an ideal gas in a closed and rigid container is increased by a factor of 4. By what factor does the pressure of the gas increase?

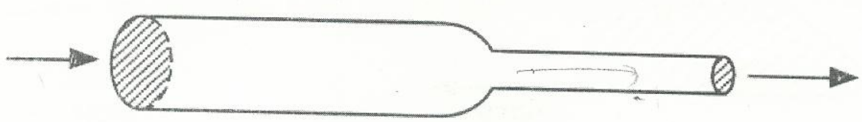
A 2
B 4
C 8
D 16

$P = \frac{1}{3}\rho c^2 \cdot P = \frac{1}{3}\frac{m}{V}c^2$

$(\frac{1}{2})\ ^2/_3$

33 The diagram below shows an ideal fluid flowing under streamline conditions in a horizontal tube.

$P = \frac{2}{3}\frac{E_k}{V}$



$P_1 = \frac{2}{3}\frac{4E_k}{V}$

$\frac{P_1}{P} = \frac{4E_k}{V} \times \frac{V}{E_k}$

Which statement about the fluid in the tube is correct?

A The density of the fluid is greater at the narrower end than at the wider end.
B Energy is conserved because work done due to the pressure difference is equal to the kinetic energy gained.
C Energy is not conserved because the pressure in the fluid decreases towards the narrower end.
D Kinetic energy is greater at the wider end than at the narrower end.

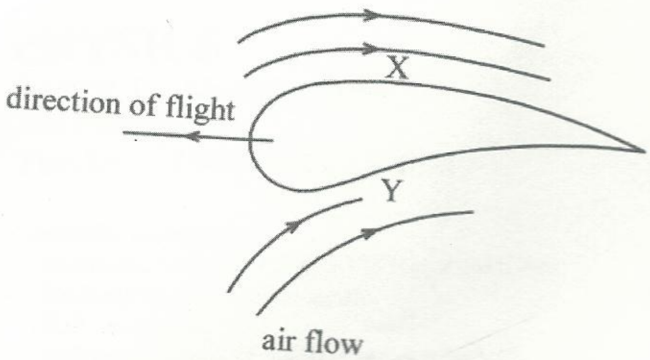
34 When heat, ΔQ is transferred to a system, its internal energy increases by ΔU and the system does external work ΔW. Which of the following expresses the first law of thermodynamics as applied to a gas?

$\Delta Q = \Delta u + \Delta W$

$\Delta u = \Delta Q - \Delta W$

A $\Delta U = \Delta Q + \Delta W$
B $\Delta U = \Delta Q - \Delta W$
C $\Delta U = \Delta W - \Delta Q$
D $\Delta U = -\Delta W - \Delta Q$

35 The diagram shows an aerofoil in flight.



Which statement is correct?

A Air moves faster at X.
B Air moves faster at Y.
C Air moves at same speed at X and Y.
D Air moves aerofoil downwards.

36 An electron is accelerated by a p.d.V. How much kinetic energy does it acquire?

A eV

VC^{-1}

B $\frac{V}{2e}$

C $\frac{eV}{2}$

D $\frac{e}{V}$

37 Two beams of light P and Q of the same wavelength fall upon the same metal surface causing photoemission of electrons. The photoelectric current produced by P is four times that produced by Q.

What is $\dfrac{\text{wave amplitude of beam } P}{\text{wave amplitude of beam } Q}$?

A $\dfrac{1}{2}$

B 2

C 4

D 16

38 Evidence for the existence and small size of the nucleus is provided by experiments in which α- particles are directed towards a gold film. Why are α- particles more preferred than β - particles?

A α- particles have higher ionising power than β - particles.
B β - particles will be repelled by the electrons of the target atoms.
C α- particles travel at higher speeds than β - particles.
D α- particles are less penetrating than β - particles.

β

39 A radioactive isotope $^{238}_{92}U$ decays to $^{234}_{92}U$ by emissions of particles. What particles are emitted?

^4_2He $^0_{-1}e$

A two alpha and one beta
B one alpha and two betas
C one alpha and three betas
D two alpha and two betas

40 An alpha particle of kinetic energy 5.4 MeV is released from a radio active material. If the activity of the material is 3.5×10^3 Bq, what is the rate of energy released by the source?

A $6.0 \times 10^{-23} Js^{-1}$
B $4.3 \times 10^{-12} Js^{-1}$
C $3.0 \times 10^{-9} Js^{-1}$
D $4.3 \times 10^{-9} Js^{-1}$

$t_{½} = \frac{\ln 2}{\lambda}$

$A = N\lambda$

$P = \frac{W}{t}$